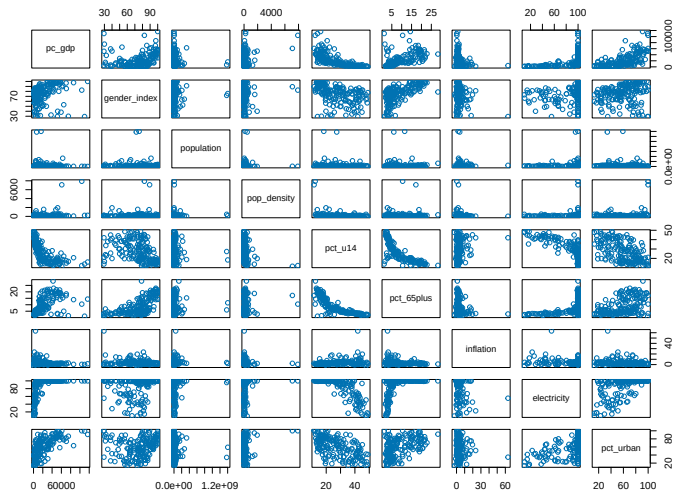




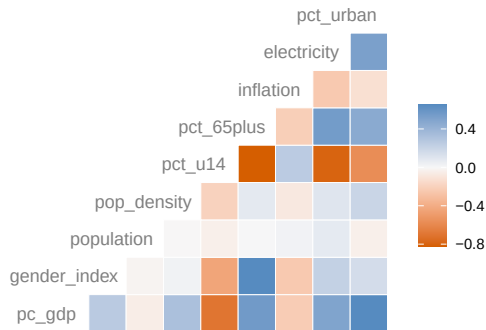
# Outline

- Principal components analysis
- $k$ -means clustering

# Exploring Relationships Between Many Variables Simultaneously



# Exploring Relationships Between Many Variables Simultaneously

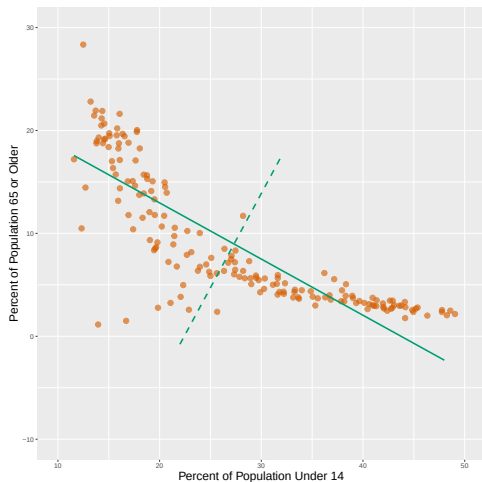


# Principal Components Analysis (PCA)

**Principal components analysis (PCA)** is a **dimension reduction** technique that summarizes the variability in a data set of  $p$  variables into  $m \leq p$  linear combinations of those variables

- Consider a data set with  $p$  variables,  $X_1, X_2, \dots, X_p$ 
  - ▶ PCA is an example of **unsupervised learning** because we analyze  $X$  variables with no  $Y$
- Each **principal component** is characterized by a set of **loadings**  $\phi_1, \dots, \phi_p$  on the variables
- The **score** is the linear combination that uses the loadings as the weights,  $\sum_{j=1}^p \phi_j x_{ij}$
- Loadings for the first principal component to maximize the variance of the score
  - ▶ Loadings for the second principal component to maximize the variance of the score, but restricting attention to loadings that are orthogonal to the first principal component
- PCA has an analytical solution based on eigendecomposition, but we won't use it

## Example: PCA with Two Variables



First principal component loadings:

- $\phi_{11} = -0.877$
- $\phi_{21} = 0.479$

First principal component direction:  $\phi_{21}/\phi_{11}$

- Passes through mean  $X_1, X_2$
- Minimizes sum of squared distances
- PC direction and OLS not identical

2<sup>nd</sup> PC direction is perpendicular to 1<sup>st</sup>

- $\phi_{12} = -0.479$
- $\phi_{22} = -0.877$

# Why Constrain the Loadings?

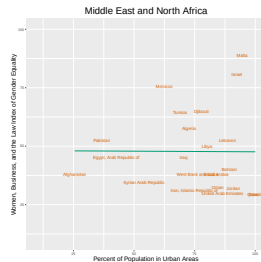
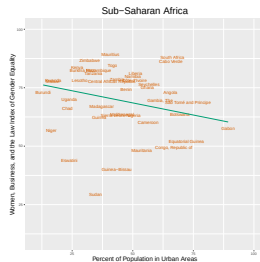
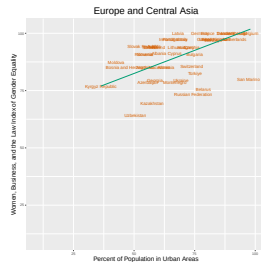
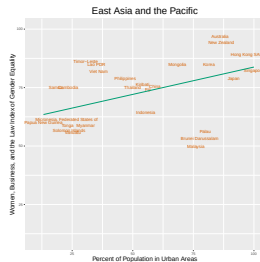
The first principal component is the set of loadings that solves:

$$\max_{\phi_1, \dots, \phi_p} \left[ \frac{1}{N} \sum_{i=1}^N \left( \sum_{j=1}^p \phi_j x_{ij} \right)^2 \right] \text{ subject to } \sum_{j=1}^p \phi_j^2 = 1$$

Why do we impose the constraint  $\sum_{j=1}^p \phi_j^2 = 1$ ?

- Multiplying loadings by a positive constant would increase the variance of the scores
- Loadings also depend on the variances of the  $X_j$  variables

# Example: Urbanization and Gender Equality Across Regions





# When to Scale $X$ Variables

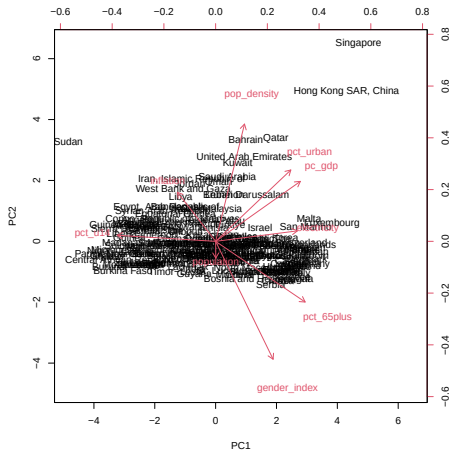
PCA over-weights high-variance variables, and variance is not unit invariant

- Common practice is to scale all variables to be mean 0, SD 1

When all variables are measured in the same units, normalization is not appropriate

- Examples: dummy variables, test items, responses in choice experiments

# Example: PCA with Nine Variables



| SD    | Prop. |
|-------|-------|
| 1.961 | 0.245 |
| 1.086 | 0.135 |
| 1.019 | 0.127 |
| 0.964 | 0.120 |
| 0.913 | 0.114 |
| 0.723 | 0.090 |
| 0.585 | 0.073 |
| 0.477 | 0.060 |
| 0.286 | 0.036 |

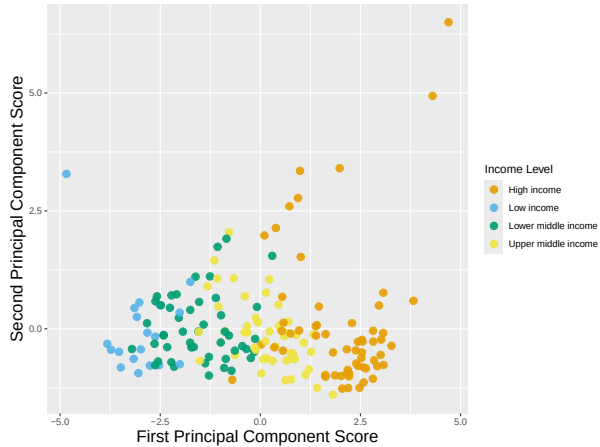
## Example: PCA with Nine Variables

| Variable     | PC1    | PC2    | PC3    | PC4    | PC5    | PC6    | PC7    | PC8    | PC9    |
|--------------|--------|--------|--------|--------|--------|--------|--------|--------|--------|
| pc_gdp       | 0.409  | 0.290  | -0.074 | 0.018  | -0.001 | -0.363 | 0.645  | 0.401  | -0.185 |
| gender_index | 0.278  | -0.569 | -0.168 | -0.128 | 0.470  | -0.164 | -0.29  | 0.463  | 0.088  |
| population   | 0.001  | -0.081 | 0.907  | -0.342 | 0.067  | -0.221 | 0.020  | 0.003  | -0.012 |
| pop_density  | 0.140  | 0.566  | -0.133 | -0.565 | 0.474  | 0.251  | -0.162 | -0.079 | -0.039 |
| pct_u14      | -0.476 | 0.029  | -0.105 | -0.088 | 0.011  | -0.277 | -0.205 | 0.169  | -0.779 |
| pct_65plus   | 0.433  | -0.293 | 0      | 0.112  | 0.239  | 0.035  | 0.138  | -0.643 | -0.472 |
| inflation    | -0.185 | 0.237  | 0.257  | 0.670  | 0.605  | 0.117  | 0.015  | 0.118  | 0      |
| electricity  | 0.395  | 0.051  | 0.221  | 0.133  | -0.320 | 0.585  | -0.258 | 0.378  | -0.347 |
| pct_urban    | 0.363  | 0.345  | 0.016  | 0.243  | -0.154 | -0.543 | -0.588 | -0.140 | 0.077  |

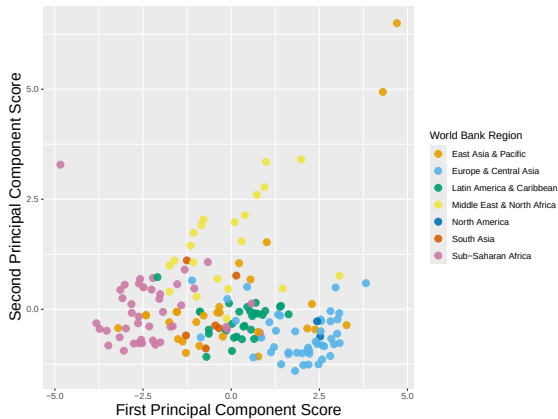
## Example: PCA with Nine Variables

| Variable     | PC1    | PC2    | PC3    | PC4    | PC5    | PC6    | PC7    | PC8    | PC9    |
|--------------|--------|--------|--------|--------|--------|--------|--------|--------|--------|
| pc_gdp       | 0.409  | 0.290  | -0.074 | 0.018  | -0.001 | -0.363 | 0.645  | 0.401  | -0.185 |
| gender_index | 0.278  | -0.569 | -0.168 | -0.128 | 0.470  | -0.164 | -0.29  | 0.463  | 0.088  |
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| pct_u14      | -0.476 | 0.029  | -0.105 | -0.088 | 0.011  | -0.277 | -0.205 | 0.169  | -0.779 |
| pct_65plus   | 0.433  | -0.293 | 0      | 0.112  | 0.239  | 0.035  | 0.138  | -0.643 | -0.472 |
| inflation    | -0.185 | 0.237  | 0.257  | 0.670  | 0.605  | 0.117  | 0.015  | 0.118  | 0      |
| electricity  | 0.395  | 0.051  | 0.221  | 0.133  | -0.320 | 0.585  | -0.258 | 0.378  | -0.347 |
| pct_urban    | 0.363  | 0.345  | 0.016  | 0.243  | -0.154 | -0.543 | -0.588 | -0.140 | 0.077  |

## Example: PCA with Nine Variables

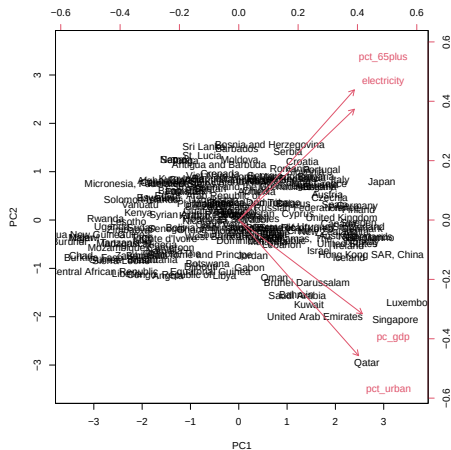


# Example: PCA with Nine Variables



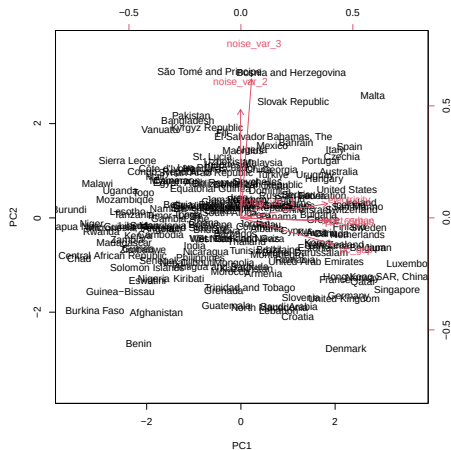
| SD    | Prop. |
|-------|-------|
| 1.961 | 0.245 |
| 1.086 | 0.135 |
| 1.019 | 0.127 |
| 0.964 | 0.120 |

# PCA with Sparse Data: Example



| Variable    | PC1   | PC2    | PC3    |
|-------------|-------|--------|--------|
| pc_gdp      | 0.521 | -0.395 | 0.322  |
| electricity | 0.487 | 0.466  | -0.702 |
| pct_65plus  | 0.487 | 0.548  | 0.593  |
| pct_urban   | 0.505 | -0.571 | -0.227 |

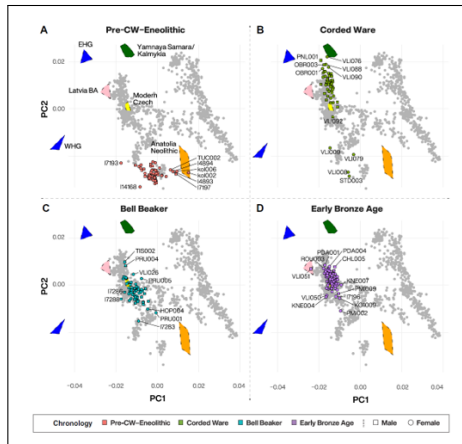
# PCA with Sparse Data: Adding Noise



| Variable    | PC1    | PC2    | PC3    |
|-------------|--------|--------|--------|
| pc-gdp      | 0.515  | -0.147 | 0.097  |
| electricity | 0.484  | 0.08   | -0.031 |
| pct_65plus  | 0.485  | -0.018 | -0.036 |
| pct_urban   | 0.497  | -0.015 | 0.192  |
| noise_var_1 | 0.12   | 0.072  | -0.787 |
| noise_var_2 | -0.002 | 0.605  | 0.498  |
| noise_var_3 | 0.06   | 0.775  | -0.291 |



# PCA in the Wild



Source: Papac et al. (2021)

# PCA: Practical Considerations

## Preprocessing:

- Which variables should be included?
- Should variables be re-scaled to have a common mean and variance?

## Interpretation:

- How many principal components to retain in subsequent analysis
- What do the different principal components represent?

# $k$ -Means Clustering

In clustering, we partition the sample into  $K$  distinct subsets or **clusters**

- Objective is to identify unobserved groups of observations
- Clustering is an **unsupervised learning** technique: no outcome variable to predict
- Groups are chosen by finding observations that look similar in terms of their  $X$  variables
- In practice, we must define “similar” in terms of some distance metric (e.g. Euclidean)

Examples of clustering in economics:

- Identifying markets and/or competitors, fields of study using text descriptions
- Identifying types of individuals, e.g. patterns of behavior in lab experiments

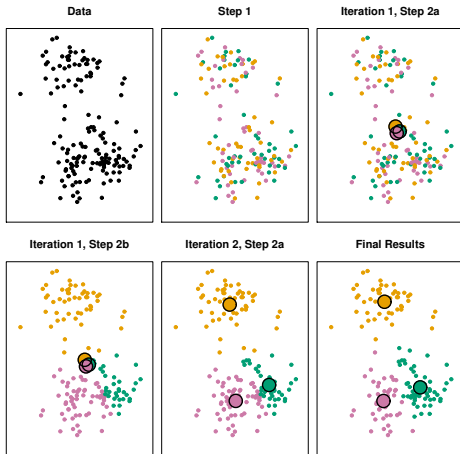
## $k$ -Means Clustering Minimizes (?) the Distance Between Observations

- Distance between  $x_a = (x_{a1}, \dots, x_{ap})$  and  $x_b = (x_{b1}, \dots, x_{bp})$ :  $d_{ab} = \sqrt{\sum_{j=1}^p (x_{aj} - x_{bj})^2}$
- Variation within cluster  $C_k$  if  $C_k$  contains  $N_{C_k}$  observations:

$$\sum_{a,b \in C} \sum_{j=1}^p (x_{aj} - x_{bj})^2 / N_{C_k} = \sum_{a,b \in C} d_{ab}^2$$

In (other) words,  $k$ -means clustering minimizes the sum of squared Euclidean distances across all pairs of points within a cluster, so that variance is between (not within) clusters

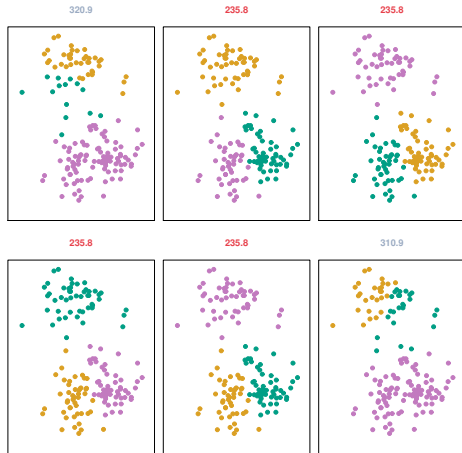
# A $k$ -Means Clustering Algorithm



Source: James et al. (2021)

1. Randomly assign initial clusters
2. Compute **centroid** of each cluster
3. Reassign clusters, mapping each observation to the nearest centroid
4. Recalculate centroids
5. Repeat Steps 3 and 4 until process converges, and no observations are reassigned at the next iteration

# $k$ -Means Clustering Identifies a Local (Not a Global) Optimum



Source: James et al. (2021)

## Example: Identifying a Global Optimum (?)

### Iteration 1: seed = 314159

| Cluster 1 ( $N = 11$ )                                                                                                                   | Cluster 2 ( $N = 162$ )                                                                                                                                                  | Cluster 3 ( $N = 2$ ) |
|------------------------------------------------------------------------------------------------------------------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-----------------------|
| Bangladesh; Brazil; Egypt, Arab Republic of; Indonesia; Japan; Mexico; Nigeria; Pakistan; Philippines; Russian Federation; United States | Afghanistan; Albania; Algeria; Angola; Antigua and Barbuda; Armenia; Australia; Austria; Azerbaijan; Bahamas, The; Bahrain; Barbados; Belarus; Belgium; Belize; Benin... | China; India          |

### Iteration 2: seed = 8675309

| Cluster 1 ( $N = 162$ )                                                                                                                                                  | Cluster 2 ( $N = 11$ )                                                                                                                   | Cluster 3 ( $N = 2$ ) |
|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------|------------------------------------------------------------------------------------------------------------------------------------------|-----------------------|
| Afghanistan; Albania; Algeria; Angola; Antigua and Barbuda; Armenia; Australia; Austria; Azerbaijan; Bahamas, The; Bahrain; Barbados; Belarus; Belgium; Belize; Benin... | Bangladesh; Brazil; Egypt, Arab Republic of; Indonesia; Japan; Mexico; Nigeria; Pakistan; Philippines; Russian Federation; United States | China; India          |

## Example: Characterizing Clusters

| Variable     | Cluster 1  | Cluster 2   | Cluster 3     |
|--------------|------------|-------------|---------------|
| pc_gdp       | 22,463     | 19,786      | 11,507        |
| gender_index | 75.21      | 69.72       | 73.75         |
| population   | 15,322,408 | 182,900,876 | 1,388,709,532 |
| pop_density  | 246.92     | 261.28      | 305.88        |
| pct_u14      | 27.37      | 27.34       | 22.83         |
| pct_65plus   | 8.95       | 9.26        | 8.77          |
| inflation    | 3.78       | 5.5         | 3.01          |
| electricity  | 84.14      | 93.52       | 97.85         |
| pct_urban    | 59.1       | 62.15       | 46.59         |



## Example: Variables the Distinguish Between Clusters

| Rank | Cluster 1    | Cluster 2    | Cluster 3    |
|------|--------------|--------------|--------------|
| 1    | pc_gdp       | pc_gdp       | population   |
| 2    | pct_urban    | pct_urban    | pop_density  |
| 3    | gender_index | electricity  | electricity  |
| 4    | pct_u14      | pct_u14      | gender_index |
| 5    | pct_65plus   | inflation    | pct_65plus   |
| 6    | inflation    | pct_65plus   | inflation    |
| 7    | electricity  | gender_index | pct_u14      |
| 8    | pop_density  | pop_density  | pct_urban    |
| 9    | population   | population   | pc_gdp       |

*Variables are ranked based on the difference in means between cluster  $j$  and the average of cluster-level means across all other clusters.*

## Example: $k$ -Means Clustering in Rescaled Data

| Cluster 1 ( $N = 41$ )                                                                                                                                                                                                                                                                                                                                                                                                                                                               | Cluster 2 ( $N = 82$ )                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                      | Cluster 3 ( $N = 52$ )                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                          |
|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Afghanistan; Angola; Benin; Burkina Faso; Burundi; Cameroon; Central African Republic; Chad; Congo, Republic of; Côte d'Ivoire; Eswatini; Gambia, The; Guinea; Guinea-Bissau; Haiti; Kenya; Lesotho; Liberia; Madagascar; Malawi; Mali; Mauritania; Micronesia, Federated States of; Mozambique; Myanmar; Namibia; Niger; Nigeria; Papua New Guinea; Rwanda; São Tomé and Príncipe; Senegal; Sierra Leone; Solomon Islands; Sudan; Tanzania; Togo; Uganda; Vanuatu; Zambia; Zimbabwe | Algeria; Antigua and Barbuda; Armenia; Azerbaijan; Bahrain; Bangladesh; Barbados; Belize; Bhutan; Bolivia, Plurinational State of; Botswana; Brazil; Brunei Darussalam; Cabo Verde; Cambodia; China; Colombia; Costa Rica; Djibouti; Dominica; Dominican Republic; Ecuador; Egypt, Arab Republic of; El Salvador; Equatorial Guinea; Fiji; Gabon; Georgia; Ghana; Grenada; Guatemala; Guyana; Honduras; India; Indonesia; Iran, Islamic Republic of; Iraq; Jamaica; Jordan; Kazakhstan; Kiribati; Kuwait; Kyrgyz Republic; Lao PDR; Lebanon; Libya; Malaysia; Maldives. . . | Albania; Australia; Austria; Bahamas, The; Belarus; Belgium; Bosnia and Herzegovina; Bulgaria; Canada; Chile; Croatia; Cyprus; Czechia; Denmark; Estonia; Finland; France; Germany; Greece; Hong Kong SAR, China; Hungary; Iceland; Ireland; Israel; Italy; Japan; Korea; Latvia; Lithuania; Luxembourg; Malta; Montenegro; Netherlands; New Zealand; North Macedonia; Norway; Poland; Portugal; Romania; Russian Federation; San Marino; Serbia; Singapore; Slovak Republic; Slovenia; Spain; Sweden; Switzerland; Ukraine; United Kingdom; United States. . . |

## Example: $k$ -Means Clustering in Rescaled Data

| Rank | Cluster 1    | Cluster 2    | Cluster 3    |
|------|--------------|--------------|--------------|
| 1    | pct_u14      | electricity  | pct_65plus   |
| 2    | inflation    | population   | pc_gdp       |
| 3    | population   | pct_urban    | gender_index |
| 4    | pop_density  | pop_density  | pct_urban    |
| 5    | gender_index | inflation    | electricity  |
| 6    | pc_gdp       | pct_u14      | pop_density  |
| 7    | pct_urban    | pc_gdp       | population   |
| 8    | pct_65plus   | gender_index | inflation    |
| 9    | electricity  | pct_65plus   | pct_u14      |

*Variables are ranked based on the difference in means between cluster  $j$  and the average of cluster-level means across all other clusters.*

# How Many Clusters?

Diagnostics for assessing the quality of a partition into clusters:

- Stability of the global maximum: does changing the seed lead to different clusters?
  - ▶ How different? We often observe clusters that are mostly but not completely stable, suggesting that there is a global maximum that is relatively flat near the peak
- How much of the total sum of squared distances between observations is between clusters?
  - ▶ Between sum of squares increases with more clusters, but by how much?
- Does the partition make sense?
  - ▶ Unsupervised learning does not usually include explicit validation, but we often have some prior knowledge of the relations between observations that we can use as a sanity check

## Cluster Diagnostics

| Clusters | Stability | Between SS | Proportion | Marginal |
|----------|-----------|------------|------------|----------|
| 3        | 4         | 646        | 0.413      |          |
| 4        | 4         | 808        | 0.516      | 0.103    |
| 5        | 3         | 963        | 0.615      | 0.099    |
| 6        | 3         | 1059       | 0.676      | 0.061    |
| 7        | 2         | 1149       | 0.734      | 0.057    |
| 8        | 2         | 1195       | 0.763      | 0.029    |
| 9        | 4         | 1233       | 0.787      | 0.024    |
| 10       | 1         |            |            |          |
| 12       | 2         | 1301       | 0.831      | 0.017    |
| 16       | 2         | 1359       | 0.868      | 0.009    |

*Variables are ranked based on the difference in means between cluster  $j$  and the average of cluster-level means across all other clusters.*

# Does the Partition Make Sense?

| Cluster 1 ( $N = 1$ ) | Cluster 2 ( $N = 82$ ) | Cluster 3 ( $N = 52$ ) |
|-----------------------|------------------------|------------------------|
| Sudan                 | China; India           | Hong Kong; Singapore   |

| Cluster 4 ( $N = 7$ )                                                               | Cluster 5 ( $N = 22$ )                                                                                                                                                                                                                                                                        | Cluster 6 ( $N = 26$ )                                                                                                                                                                                                                                                                                     |
|-------------------------------------------------------------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Bahrain; Brunei Darussalam; Kuwait; Oman; Qatar; Saudi Arabia; United Arab Emirates | Afghanistan; Algeria; Bangladesh; Botswana; Djibouti; Egypt, Arab Republic of; Equatorial Guinea; Eswatini; Gabon; Ghana; Indonesia; Iran, Islamic Republic of; Iraq; Jordan; Lebanon; Libya; Malaysia; Pakistan; São Tomé and Príncipe; Syrian Arab Republic; Uzbekistan; West Bank and Gaza | Antigua and Barbuda; Barbados; Belize; Bhutan; Cambodia; Fiji; Grenada; Guatemala; Guyana; Honduras; Kiribati; Kyrgyz Republic; Lao PDR; Maldives; Micronesia, Federated States of; Myanmar; Nepal; Nicaragua; Philippines; Samoa; Sri Lanka; St. Kitts and Nevis; St. Lucia; Timor-Leste; Tonga; Viet Nam |

## Does the Partition Make Sense?

| Cluster 7 ( $N = 35$ )                                                                                                                                                                                                                                                                                                                                                                | Cluster 8 ( $N = 39$ )                                                                                                                                                                                                                                                                                                                                                                   | Cluster 9 ( $N = 41$ )                                                                                                                                                                                                                                                                                                                                                                                                                                                                                              |
|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Angola; Benin; Burkina Faso; Burundi; Cameroon; Central African Republic; Chad; Congo, Republic of; Côte d'Ivoire; Gambia, The; Guinea; Guinea-Bissau; Haiti; Kenya; Lesotho; Liberia; Madagascar; Malawi; Mali; Mauritania; Mozambique; Namibia; Niger; Nigeria; Papua New Guinea; Rwanda; Senegal; Sierra Leone; Solomon Islands; Tanzania; Togo; Uganda; Vanuatu; Zambia; Zimbabwe | Australia; Austria; Belgium; Bulgaria; Canada; Croatia; Cyprus; Czechia; Denmark; Estonia; Finland; France; Germany; Greece; Hungary; Iceland; Ireland; Italy; Japan; Korea; Latvia; Lithuania; Luxembourg; Malta; Netherlands; New Zealand; Norway; Poland; Portugal; Romania; San Marino; Serbia; Slovak Republic; Slovenia; Spain; Sweden; Switzerland; United Kingdom; United States | Albania; Armenia; Azerbaijan; Bahamas, The; Belarus; Bolivia, Plurinational State of; Bosnia and Herzegovina; Brazil; Cabo Verde; Chile; Colombia; Costa Rica; Dominica; Dominican Republic; Ecuador; El Salvador; Georgia; Israel; Jamaica; Kazakhstan; Mauritius; Mexico; Moldova; Mongolia; Montenegro; Morocco; North Macedonia; Palau; Panama; Paraguay; Peru; Russian Federation; Seychelles; South Africa; St. Vincent and the Grenadines; Thailand; Trinidad and Tobago; Tunisia; Türkiye; Ukraine; Uruguay |

# Does the Partition Make Sense?

| Cluster 1 ( $N = 1$ )                                     | Cluster 2 ( $N = 82$ ) | Cluster 3 ( $N = 52$ )                                      |
|-----------------------------------------------------------|------------------------|-------------------------------------------------------------|
| Sudan                                                     | China; India           | Hong Kong; Singapore                                        |
| inflation +<br>gender equality –<br>population under 14 + | population +           | population density +<br>GDP per capita +<br>percent urban + |



# Does the Partition Make Sense?

| Cluster 4 ( $N = 7$ )                                                               | Cluster 5 ( $N = 22$ )                                                                                                                                                                                                                                                                        | Cluster 6 ( $N = 26$ )                                                                                                                                                                                                                                                                                     |
|-------------------------------------------------------------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Bahrain; Brunei Darussalam; Kuwait; Oman; Qatar; Saudi Arabia; United Arab Emirates | Afghanistan; Algeria; Bangladesh; Botswana; Djibouti; Egypt, Arab Republic of; Equatorial Guinea; Eswatini; Gabon; Ghana; Indonesia; Iran, Islamic Republic of; Iraq; Jordan; Lebanon; Libya; Malaysia; Pakistan; São Tomé and Príncipe; Syrian Arab Republic; Uzbekistan; West Bank and Gaza | Antigua and Barbuda; Barbados; Belize; Bhutan; Cambodia; Fiji; Grenada; Guatemala; Guyana; Honduras; Kiribati; Kyrgyz Republic; Lao PDR; Maldives; Micronesia, Federated States of; Myanmar; Nepal; Nicaragua; Philippines; Samoa; Sri Lanka; St. Kitts and Nevis; St. Lucia; Timor-Leste; Tonga; Viet Nam |
| GDP per capita +<br>gender equality –<br>percent urban +                            |                                                                                                                                                                                                                                                                                               |                                                                                                                                                                                                                                                                                                            |

# Does the Partition Make Sense?

| Cluster 7 ( $N = 35$ )                                                                                                                                                                                                                                                                                                                                                                | Cluster 8 ( $N = 39$ )                                                                                                                                                                                                                                                                                                                                                                   | Cluster 9 ( $N = 41$ )                                                                                                                                                                                                                                                                                                                                                                                              |
|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Angola; Benin; Burkina Faso; Burundi; Cameroon; Central African Republic; Chad; Congo, Republic of; Côte d'Ivoire; Gambia, The; Guinea; Guinea-Bissau; Haiti; Kenya; Lesotho; Liberia; Madagascar; Malawi; Mali; Mauritania; Mozambique; Namibia; Niger; Nigeria; Papua New Guinea; Rwanda; Senegal; Sierra Leone; Solomon Islands; Tanzania; Togo; Uganda; Vanuatu; Zambia; Zimbabwe | Australia; Austria; Belgium; Bulgaria; Canada; Croatia; Cyprus; Czechia; Denmark; Estonia; Finland; France; Germany; Greece; Hungary; Iceland; Ireland; Italy; Japan; Korea; Latvia; Lithuania; Luxembourg; Malta; Netherlands; New Zealand; Norway; Poland; Portugal; Romania; San Marino; Serbia; Slovak Republic; Slovenia; Spain; Sweden; Switzerland; United Kingdom; United States | Albania; Armenia; Azerbaijan; Bahamas, The; Belarus; Bolivia, Plurinational State of; Bosnia and Herzegovina; Brazil; Cabo Verde; Chile; Colombia; Costa Rica; Dominica; Dominican Republic; Ecuador; El Salvador; Georgia; Israel; Jamaica; Kazakhstan; Mauritius; Mexico; Moldova; Mongolia; Montenegro; Morocco; North Macedonia; Palau; Panama; Paraguay; Peru; Russian Federation; Seychelles; South Africa... |
| population under 14 +<br>electricity –                                                                                                                                                                                                                                                                                                                                                | population 65 and over +<br>gender equality +                                                                                                                                                                                                                                                                                                                                            |                                                                                                                                                                                                                                                                                                                                                                                                                     |

# $k$ -Means Clustering: Practical Considerations

- Should  $X$  variables be normalized or not?
- Which  $X$  variables should be included?
- How many clusters? Should they be small or large?
  - ▶ Do the clusters identify meaningful groups?
- How many random starts are needed to avoid local maxima? How robust are the clusters?
  - ▶ Check for (unexpected) clusters that are very small, exclude outliers or adapt preprocessing
  - ▶ Clustering can be very sensitive to small changes in sample construction (so beware)